

Problem 3.2

Solution:

(i) Subproblems:

Define $C(i, j, m)$ to be the minimum cost over all binary trees T for the subsequence $\langle a_i, a_{i+1}, \dots, a_j \rangle$ with root a_m . Want $\min_{1 \leq m \leq n} C(1, n, m)$.

(ii) Recursive formula:

$$C(i, j, m) = \min_{i \leq l < m < r \leq j} \{C(i, m-1, l) + C(m+1, j, r) + |a_l - a_m| + |a_r - a_m|\}$$

Base case: $C(i, i+1, m) = |a_i - a_{i+1}|$ for $1 \leq i < n$.

(iii) Evaluation order:

Increase j from 1 to n , for each j , decrease i from j to 1, for each i, j , evaluate m from i to j .

(iv) Time and space complexity:

Iteration takes $O(n^3)$ steps, each step takes $O(n)$ time to evaluate.

Total time complexity: $O(n^4)$.

Total space complexity: $O(n^3)$.